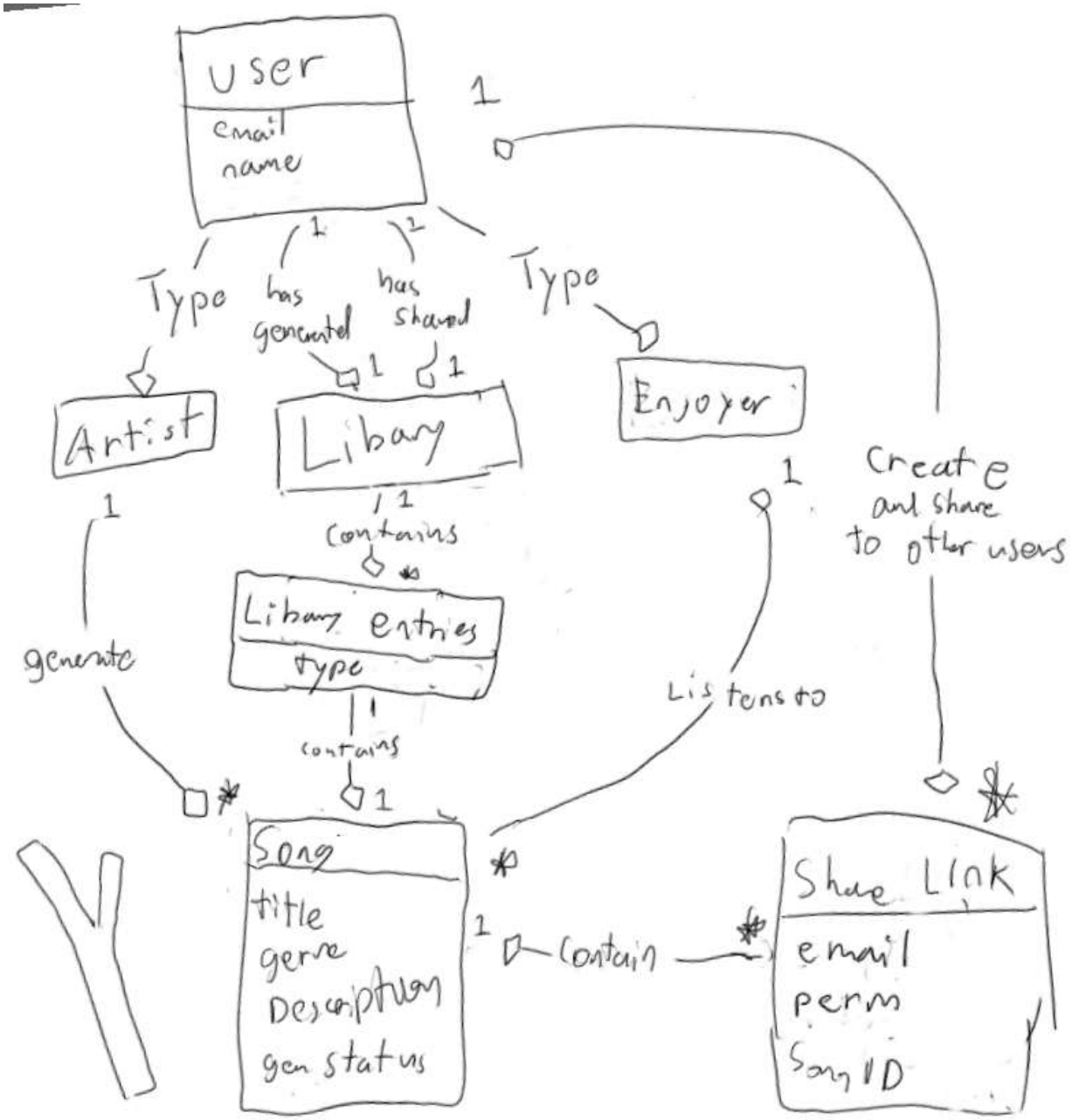
Chithara AI Music Generator

Company name

Domain Diagram

Author: 6610545316 Taam Paramee

Hand Drawn model



(Leak for authenticity)

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Description

The domain model represents a Music Generation AI application with two Conceptual user types: **Artist** and **Enjoyer**, both are conceptual types and not physical in the system but modeled for better understanding of user groups.

- An **Artist** are users that mainly generate, manage, and share songs. Each **Song** is owned by exactly one Artist and contains essential metadata such as title, description, genre, and generation status.
- An **Enjoyer** are users that mainly generate songs but can access songs shared with them, listen to them, and download them if permitted.

Each User has two conceptual libraries:

- A Generated Library (for owned songs)
- A Shared Library (for songs shared with the user)

Songs are shared using a **ShareLink**, which controls access through visibility settings **ShareLink.Type:** (public or email-restricted) and with **ShareLink.Permission:** (download permissions, Share forwarding permissions). Business rules ensure that only the owning Artist can manage a song and that sharing permissions are enforced. Users that share forward can only share the same link forward and cannot change or create a new sharelink and set of permissions.

Enumerations / Value Types

- **User.Email** Email value type
- **User.Name** String Value Type
- **ShareLink.Email** Email value type
- **ShareLink.Type** = { PUBLIC, EMAIL }
- **ShareLink.Permissions** = [VIEW,DOWNLOAD,SHARE] Array Enum
- **LibraryEnteries.Type** = { GENERATED, SHARED}
- **Song.Title** String value type
- **Song.Genre** = {POP, ROCK, JAZZ ..etc}
- **Song.Description** String value type
- **Song.GenerationStatus** = { QUEUED, RUNNING, SUCCESS, FAILED }

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Key Constraints

1. A Song must belong to exactly one Artist.
2. Only song Owners can Generate Share links and set permissions.
3. A Song must always have a defined generationStatus.
4. If **ShareLink.Type** = EMAIL , only emails in allowedEmails may access.
5. View Download and Forwarding are allowed if **ShareLink.Permissions** contains: (VIEW,DOWNLOAD,SHARE) respectively
6. Shared songs appear in the user's sharedLibrary.
7. Only the owning Artist can delete a Song.

Traceability Table

Domain Class	User Stories	System Features	Use Cases	Functional Requirements
User	US1, US2	SF1	UC3, UC4	FR1, FR11
Artist	US3–US11, US18	SF2, SF3, SF4, F5, F6	UC1, UC5, UC7, UC8, UC12, UC13	FR2, FR3, FR4, FR5, FR6, FR7, FR8, FR10
Enjoyer	US12–US15	F5, F7	UC2, UC6, UC9, UC10, UC11	FR4, FR7, FR9, FR12
Song	US3–US6, US8, US11–US13	SF2, SF3, SF4, F5	UC1, UC2, UC5, UC7, UC8, UC11	FR2, FR3, FR5, FR6, FR7, FR10, FR12
Library	US6, US14	SF3, F7	UC5, UC9	FR3, FR9
LibraryEntry	US14, US15	F7	UC9, UC10	FR9
ShareLink	US8–US10, US12–US13	F5, F6	UC2, UC12, UC13	FR6, FR7, FR8, FR12

AI was used to help with Traceability mapping.